

GED Math Unit: Data, Statistics, and Probability

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Objectives

- Identify data types: categorical vs. numerical (discrete/continuous).
- Organize data using frequency and relative-frequency tables.
- Understand populations, samples, and basic sampling methods.
- Recognize sources of bias and good survey practices.

- **Categorical data:** labels or groups (e.g., color, brand).
- **Numerical data:** numbers that measure/count.
 - **Discrete:** counts (0,1,2,...).
 - **Continuous:** measured on a scale (height, time).
- **Frequency table:** lists categories/values with counts.
- **Relative frequency:** frequency divided by total.
- **Population:** entire group; **sample:** subset.

Sampling Methods & Bias

- **Simple random sample:** every member equally likely.
- **Systematic:** every k th individual.
- **Stratified:** split into subgroups, sample from each.
- **Cluster:** randomly select groups, survey all within.
- **Convenience/Voluntary response:** often biased.
- **Bias** occurs when sample or questions systematically favor an outcome.

Worked Example 1

Example. Classify each: (a) brand of phone; (b) number of siblings; (c) commute time.

Worked Solution.

- ① (a) **Categorical** (brand is a label).
- ② (b) **Numerical discrete** (counts: $0, 1, 2, \dots$).
- ③ (c) **Numerical continuous** (time measured on a scale).

Worked Example 2

Example. Create a frequency and relative-frequency table for colors: R, B, B, G, B, R, G (R=Red, B=Blue, G=Green).

Worked Solution.

- 1 Count: R:2, B:3, G:2 (total $n = 7$).
- 2 Relative freq: R: $2/7 \approx 0.286$, B: $3/7 \approx 0.429$, G: $2/7 \approx 0.286$.

Worked Example 3

Example. A school of 1200 students wants average screen time. Suggest a sampling plan that limits bias.

Worked Solution.

- 1 Use a **stratified random sample** by grade: 9–12.
- 2 From each grade, randomly select proportional numbers (e.g., 25 from each if grades are similar size).
- 3 Ensure **neutral wording** and anonymous responses to reduce response bias.

Worked Example 4

Example. In a production line, survey every 10th item. What method is this and a potential risk?

Worked Solution.

- 1 **Systematic sampling** with $k = 10$.
- 2 Risk: if defects occur periodically every 10 items, bias may occur.

Worked Example 5

Example. A website poll asks, “Do you agree that students spend too much time on phones?” Identify the issue.

Worked Solution.

- ① **Voluntary response** (self-selection) and **leading question** (“too much”) introduce bias.

Objectives

- Construct and interpret bar charts (categorical) and histograms (numerical).
- Read dot plots and pie charts; compare distributions.
- Describe shape: symmetric, skewed, unimodal/multimodal, gaps, outliers.

- **Bar chart:** separated bars for categories.
- **Histogram:** contiguous bars for bins of numerical data.
- **Dot plot:** shows each observation as a dot over a number line.
- **Pie chart:** sector angle $\propto$ category fraction.
- Look for **center, spread, shape, outliers**.

Worked Example 1

Example. Categorical data of car colors: Black 18, White 12, Gray 10, Red 5. What display?

Worked Solution.

- 1 Use a **bar chart** (categories).
- 2 Heights proportional to counts: 18, 12, 10, 5.

Worked Example 2

Example. Create histogram bins for commute times (min) 5,7,9,12,14,16,20,21,22 with bin width 5.

Worked Solution.

- 1 Bins: $[5,10):3$ (5,7,9); $[10,15):2$ (12,14); $[15,20):1$ (16); $[20,25):3$ (20,21,22).
- 2 Bars touch because continuous scale.

Worked Example 3

Example. A dot plot clusters around 10–12 with a few at 3 and 20. Describe shape.

Worked Solution.

- 1 **Unimodal** center near 11; potential **outliers** at 3 and 20; mild **skewness** depending on tail lengths.

Worked Example 4

Example. Pie chart sectors for categories A:30, B:45, C:25. Compute angles.

Worked Solution.

- 1 Total = 100. Angles = fraction $\times 360^\circ$.
- 2 A: $0.30 \cdot 360^\circ = 108^\circ$, B: 162° , C: 90° .

Worked Example 5

Example. Two histograms: Set X tightly around 50; Set Y spread 30–70. Who has greater variability?

Worked Solution.

- 1 Set Y shows greater spread; Set X is more consistent.

Objectives

- Compute mean, median, mode, and range.
- Find quartiles, interquartile range (IQR), and identify outliers.
- Understand mean absolute deviation (MAD) conceptually.

Key Formulas

- Mean: $\bar{x} = \frac{\sum x_i}{n}$.
- Median: middle after sort (or average of two middle values).
- IQR: $Q_3 - Q_1$. Outlier rule: below $Q_1 - 1.5\text{IQR}$ or above $Q_3 + 1.5\text{IQR}$.
- Range: $\max - \min$. MAD: average of $|x_i - \bar{x}|$.

Worked Example 1

Example. Data: 4, 7, 8, 9, 10. Find mean, median, mode, range.

Worked Solution.

- 1 Mean: $(4 + 7 + 8 + 9 + 10)/5 = 38/5 = 7.6$.
- 2 Median: middle is 8.
- 3 Mode: none (all unique).
- 4 Range: $10 - 4 = 6$.

Worked Example 2

Example. Data: 1, 2, 2, 5, 7, 9. Find median.

Worked Solution.

- ① Even $n = 6$: middle two are 2 and 5. Median = $(2 + 5)/2 = 3.5$.

Worked Example 3

Example. Find Q_1 , Q_3 , and IQR for data: 3,4,5,7,8,10,12,13,15.

Worked Solution.

- 1 Sorted already, $n = 9$. Median is 8 (5th value).
- 2 Lower half: 3,4,5,7 $\Rightarrow Q_1$ median = $(4 + 5)/2 = 4.5$.
- 3 Upper half: 10,12,13,15 $\Rightarrow Q_3$ median = $(12 + 13)/2 = 12.5$.
- 4 IQR = $12.5 - 4.5 = 8$.

Worked Example 4

Example. Using the IQR rule, are there outliers in data: 2,3,4,6,7,7,8,50?

Worked Solution.

- ① Sorted: 2,3,4,6,7,7,8,50; $n = 8$.
- ② Median = $(6 + 7)/2 = 6.5$. Lower: 2,3,4,6 $\Rightarrow Q_1 = (3 + 4)/2 = 3.5$.
- ③ Upper: 7,7,8,50 $\Rightarrow Q_3 = (7 + 8)/2 = 7.5$.
- ④ IQR = $7.5 - 3.5 = 4$. Fences: $3.5 - 1.5(4) = -2.5$ and $7.5 + 1.5(4) = 13.5$.
- ⑤ $50 > 13.5 \Rightarrow$ **50 is an outlier.**

Worked Example 5

Example. Compute the mean and MAD for data: 2, 4, 6, 8.

Worked Solution.

- 1 Mean $\bar{x} = (2 + 4 + 6 + 8)/4 = 5$.
- 2 Deviations: $|2 - 5| = 3$, $|4 - 5| = 1$, $|6 - 5| = 1$, $|8 - 5| = 3$.
- 3 MAD = $(3 + 1 + 1 + 3)/4 = 2$.

Objectives

- Find the five-number summary: $\min$, Q_1 , Median, Q_3 , $\max$.
- Draw and interpret a box-and-whisker plot.
- Compare two boxplots to discuss center and spread.

Key Ideas

- Box spans from Q_1 to Q_3 ; line at the median.
- Whiskers extend to min/max that are not outliers.
- Outliers plotted as individual points.
- IQR measures the middle 50% spread.

Worked Example 1

Example. Data: 1,3,4,4,6,7,9,11. Find five-number summary.

Worked Solution.

- ① $\min = 1, \max = 11, n = 8.$
- ② $\text{Median} = (4 + 6)/2 = 5.$
- ③ Lower half: 1,3,4,4 $\Rightarrow Q_1 = (3 + 4)/2 = 3.5.$
- ④ Upper half: 6,7,9,11 $\Rightarrow Q_3 = (7 + 9)/2 = 8.$

Worked Example 2

Example. Sketch the boxplot using the summary from Example 1.

Worked Solution.

- 1 Draw number line from 1 to 11.
- 2 Box from $Q_1 = 3.5$ to $Q_3 = 8$ with median at 5.
- 3 Whiskers to 1 and 11 (no outliers by IQR rule checked below).

Worked Example 3

Example. Check for outliers using IQR in Example 1.

Worked Solution.

- 1 IQR = $8 - 3.5 = 4.5$. Fences: $3.5 - 1.5(4.5) = -3.25$, $8 + 1.5(4.5) = 14.75$.
- 2 All data between 1 and 11, so none are outliers.

Worked Example 4

Example. Compare two boxplots: Class A median 75, IQR 10; Class B median 72, IQR 20.
Who is more consistent?

Worked Solution.

- 1 Class A has smaller IQR (10 vs 20) $\Rightarrow$ more consistent (less spread).
- 2 Class A also has slightly higher center.

Worked Example 5

Example. Given a boxplot with lower whisker at 20, $Q_1 = 30$, median 40, $Q_3 = 50$, upper whisker at 65, estimate the percentage of data between 30 and 50.

Worked Solution.

- 1 The box from Q_1 to Q_3 contains the middle 50% of data.
- 2 Therefore, about **50%** of data lie between 30 and 50.

Objectives

- Compute simple probabilities from equally likely outcomes.
- Understand complements, independent vs. dependent events.
- Use formulas for compound probability (AND/OR).

Key Formulas

- $P(\text{event}) = \frac{\text{favorable}}{\text{total}}$ (equally likely).
- Complement: $P(A^c) = 1 - P(A)$.
- Independent: $P(A \cap B) = P(A)P(B)$.
- General OR: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Worked Example 1

Example. Roll a fair die. $P(\text{even})$?

Worked Solution.

- 1 Favorable outcomes: $\{2,4,6\}$ (3 outcomes), total 6.
- 2 $P(\text{even}) = 3/6 = 1/2$.

Worked Example 2

Example. Draw one card from a standard 52-card deck. $P(\text{heart})$ and $P(\text{not heart})$?

Worked Solution.

- 1 Hearts: 13 of 52 $\Rightarrow P(\text{heart}) = 13/52 = 1/4$.
- 2 Complement: $P(\text{not heart}) = 1 - 1/4 = 3/4$.

Worked Example 3

Example. Two independent coin flips. $P(\text{at least one head})$?

Worked Solution.

- 1 Complement: $P(\text{no heads}) = P(TT) = 1/4$.
- 2 So $P(\geq 1 \text{ H}) = 1 - 1/4 = 3/4$.

Worked Example 4

Example. Bag with 5 red and 3 blue. Draw 2 without replacement. $P(\text{both red})$?

Worked Solution.

- 1 First red: $5/8$. Second red given first red: $4/7$.
- 2 $P = (5/8)(4/7) = 20/56 = 5/14 \approx 0.357$.

Worked Example 5

Example. In a class, $P(A) = 0.4$, $P(B) = 0.3$, $P(A \cap B) = 0.1$. Find $P(A \cup B)$.

Worked Solution.

$$\textcircled{1} P(A \cup B) = 0.4 + 0.3 - 0.1 = 0.6.$$

Objectives

- Read and compute probabilities from two-way (contingency) tables.
- Compute conditional probability $P(A \mid B)$.
- Use permutations/combinations for counting.

Key Ideas & Formulas

- Two-way table organizes joint counts. Marginals on edges.
- Conditional probability: $P(A \mid B) = \frac{P(A \cap B)}{P(B)}$.
- Permutations (order matters): ${}_nP_r = \frac{n!}{(n-r)!}$.
- Combinations (order not): ${}_nC_r = \frac{n!}{r!(n-r)!}$.

Worked Example 1

Example. Two-way table of 100 students:

	Like Math	Neutral	Dislike	Total
Freshman	18	7	5	30
Sophomore	15	10	5	30
Upperclass	20	12	8	40
Total	53	29	18	100

Worked Solution.

- 1 $P(\text{Like}) = 53/100 = 0.53.$
- 2 $P(\text{Freshman}) = 30/100 = 0.30.$

Worked Example 2

Example. From the table, find $P(\text{Like} \mid \text{Freshman})$.

Worked Solution.

- 1 Within Freshmen, total = 30, Like = 18.
- 2 $P(\text{Like} \mid \text{Freshman}) = 18/30 = 0.6$.

Worked Example 3

Example. Are *Freshman* and *Like* independent?

Worked Solution.

- 1 Check if $P(\text{Like} \mid \text{Freshman}) = P(\text{Like})$.
- 2 $0.6 \neq 0.53 \Rightarrow$ not independent.

Worked Example 4

Example. Passwords with letters A–Z (26) and digits 0–9 (10). How many 4-character codes with no repeats if (a) letters only; (b) letters or digits?

Worked Solution.

- ① (a) Permutations without repeat: $26 \cdot 25 \cdot 24 \cdot 23 = {}_{26}P_4$.
- ② (b) From 36 symbols: $36 \cdot 35 \cdot 34 \cdot 33 = {}_{36}P_4$.

Worked Example 5

Example. A committee of 3 from 8 people. How many ways?

Worked Solution.

① Order does not matter $\Rightarrow$ combinations: ${}_8C_3 = \frac{8!}{3!5!} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56.$

Lesson 1 Practice (Assign)

- Identify type: (a) shoe size; (b) city of birth; (c) temperature.
- Make a relative-frequency table for: A,A,B,C,A,B,C,C.
- Propose an unbiased sampling plan to estimate average sleep hours.

Lesson 2 Practice (Assign)

- Choose an appropriate display for survey data of favorite genres; justify.
- Given a histogram sketch, describe center, spread, shape, outliers.

Lesson 3 Practice (Assign)

- Compute mean/median/mode/range for: 5,7,7,8,9,10.
- Find Q_1 , Q_3 , IQR for: 2,3,4,6,7,8,12,14.
- Use IQR rule to test for outliers in: 1,2,2,3,4,80.

Lesson 4 Practice (Assign)

- Draw a boxplot for: 12,14,15,16,18,21,22,24,30.
- Compare two boxplots with same median but different IQRs.

Lesson 5 Practice (Assign)

- Deck of 52: $P(\text{face card})$; $P(\text{not diamond})$.
- Urn: 4 green, 6 yellow. Draw 2 without replacement: $P(\text{one of each})$.

Lesson 6 Practice (Assign)

- From a two-way table, compute a conditional probability and decide independence.
- Locker combinations: digits 0–9, 3 digits, repetition allowed. How many?

Unit Summary

- Choose displays that match data type; describe distributions.
- Mean/median/IQR tell center and spread; use IQR rule for outliers.
- Probability basics: complements, AND/OR, independence vs dependence.
- Conditional probability, two-way tables, and counting principles.

Quick Exit Ticket

- 1 Data: 2,3,3,5,7,9. Find median and IQR.
- 2 Bag: 3 red, 2 blue. Draw 2 without replacement. $P(\text{both blue})$?
- 3 True/False: If $P(A | B) = P(A)$, then A and B are independent.

Questions?